HARRIS TEETER · PREPARED FOODS · PROPOSAL FOR LEADERSHIP

Fresh Forecast

Measure, predict and reduce prepared-food waste,
one store at a time, starting with a 90-day pilot.

- Concept validated in a full-year simulation
- Built on zero company data
- September 2026

THE ONE-SLIDE VERSION

We can stop guessing how much to make

Bakery, pizza, hot foods and fresh cases are produced against gut-feel par sheets. Pars cannot see weather, holidays or trend, so they pad, and the padding becomes waste. A demand forecast can see all of those a day ahead.

A working forecasting system already exists. It was built and tested on a fully synthetic store so that no data-permission or IP question had to be answered first. The ask is one pilot store, read-only data access, and a modest number of hours.

−37% waste in the simulated pilot year with product availability held at today's level

81% of the improvement that perfect knowledge of demand would allow

10–15% the waste reduction this proposal actually promises for a real 90-day pilot

Simulation figures come from a synthetic three-year store, held-out year 2025. They prove the method. The real dollar figure comes from the pilot's measured baseline.

THE PROBLEM

A par sheet is one number. Demand is not.

Make too much and we eat the cost of goods. Make too little and we lose the margin and disappoint the customer at an empty hot bar at 6 pm.

Pars solve this with a flat safety pad. But demand swings with the weekday, the weather, the season and the calendar, and all of those are known the day before.

Demand index by weekday and weather in the simulated store. 1.00 is an average day. The spread between a snowy Monday and a sunny Friday is almost 2×, and a single par number has to cover both.

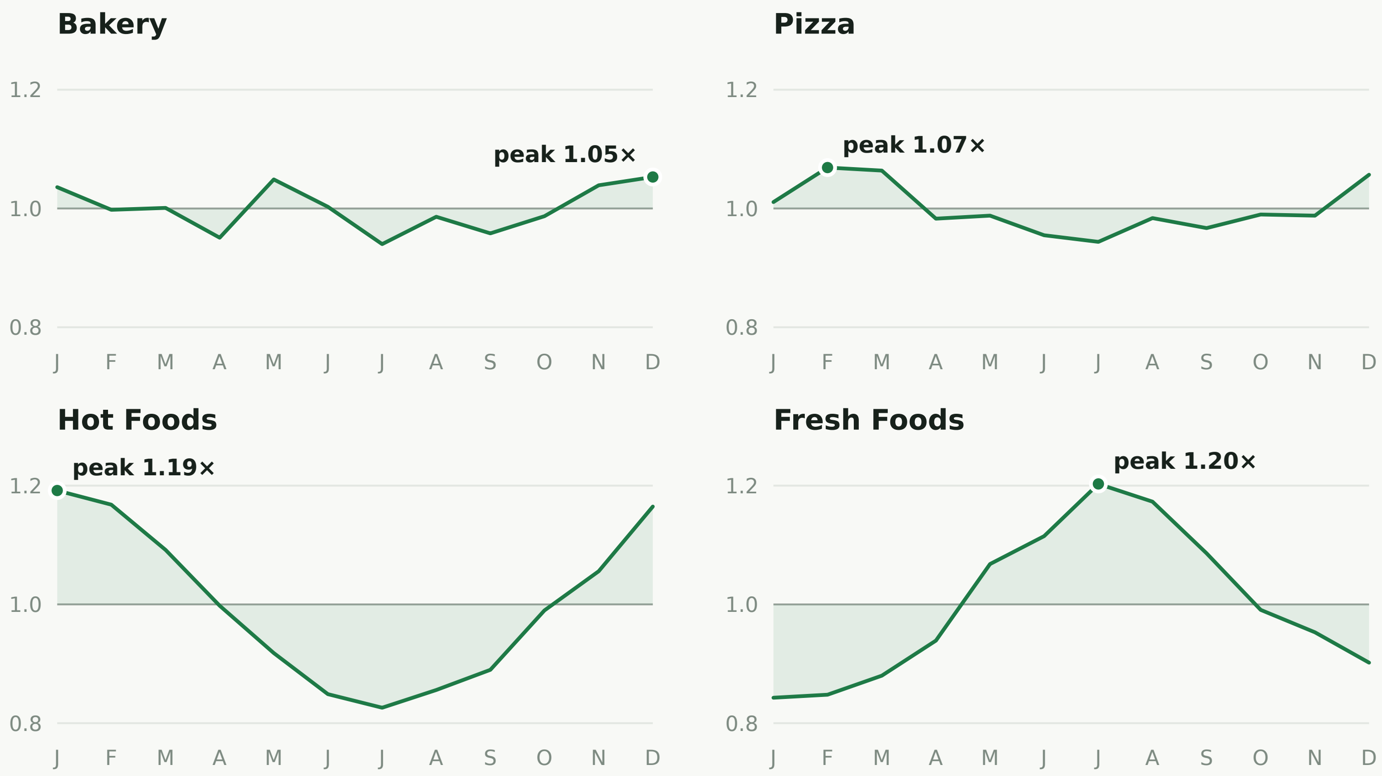
	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Sunny	1.02	1.00	1.05	1.05	1.22	1.08	0.99
Cloudy	0.99	0.94	1.00	1.03	1.16	1.04	0.95
Rain	0.87	0.85	0.91	0.90	1.05	0.92	0.84
Snow	0.64	n/a	n/a	n/a	0.60	n/a	0.61

THE PROBLEM, CONTINUED

Each department has its own year

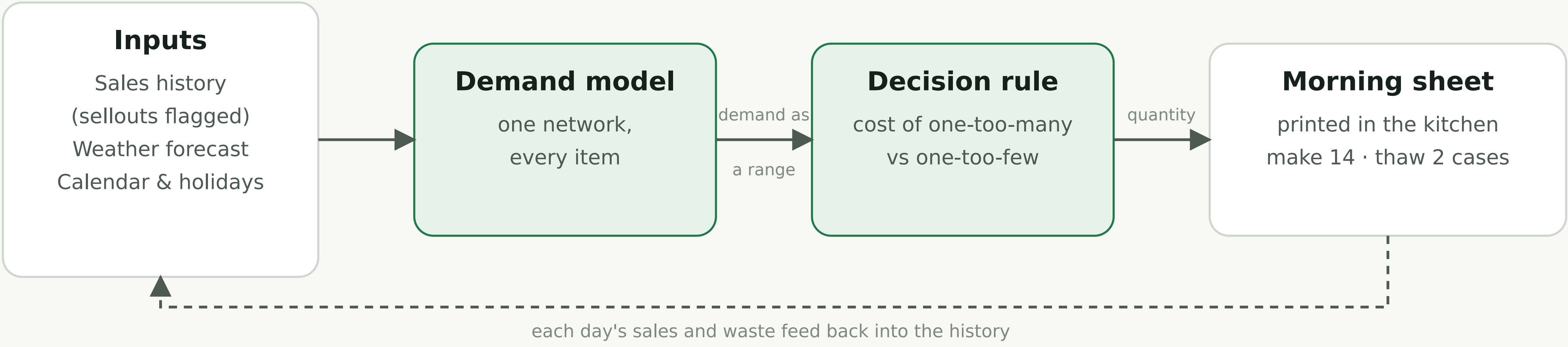
Hot foods peak in winter, fresh foods in summer, bakery and pizza around the holidays. A trailing average lags each of these turns by weeks. A forecast that knows the calendar turns with them.

Monthly demand index by department in the simulated store, relative to each department's own average. The shaded area is the departure from 1.00.



WHAT ALREADY EXISTS

The whole system, end to end



No integration needed for a pilot. The inputs are a sales export, a public weather forecast and a calendar. The output is a printed sheet in the kitchen each morning.

One model, every item. A compact neural network of the family used industrially for retail demand forecasting, forecasting a demand range rather than a point.

A decision rule, not a guess. Produce where the cost of one wasted unit equals the cost of one missed sale. Cheap, high-margin items pad; expensive, low-margin items run lean.

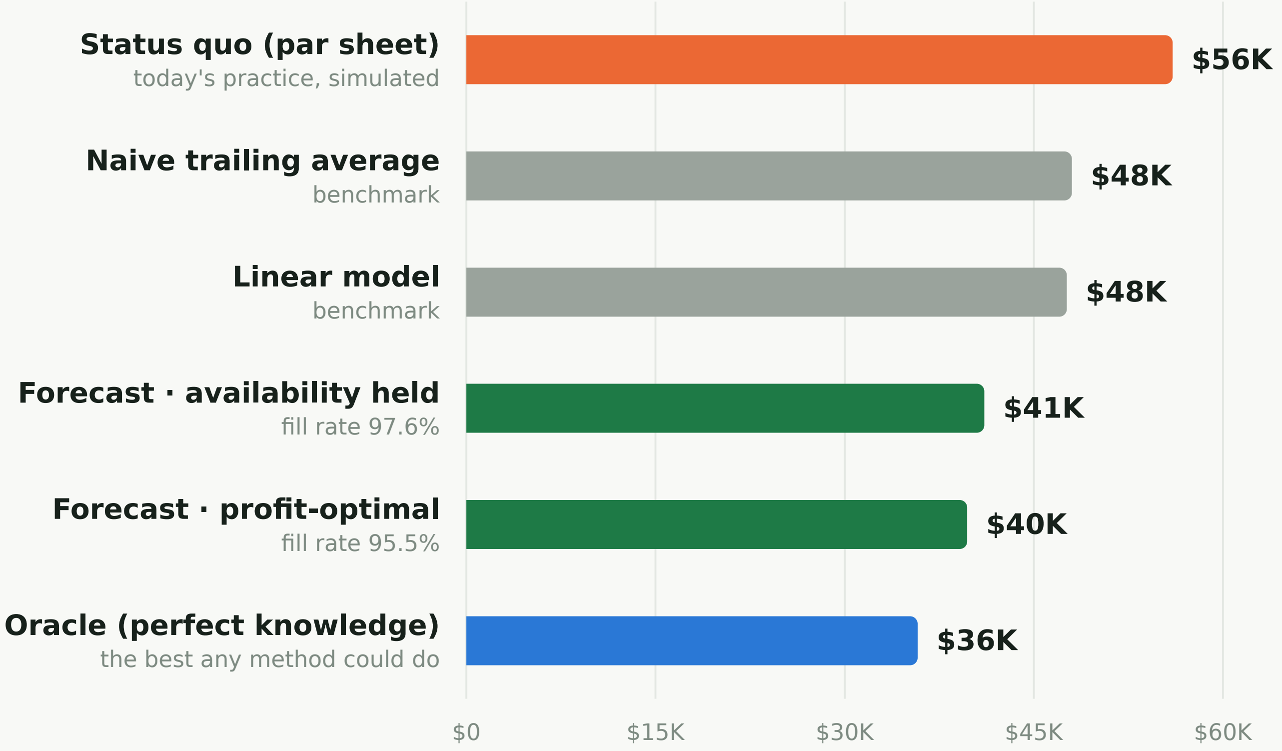
PROOF OF CONCEPT

A year replayed: the forecast closes 81% of the gap to perfect knowledge

The model trained on two simulated years, seeing only what a real store sees, then ran in shadow against the untouched third year. For every day and item it recommended a quantity; the simulation's true demand settled what sold, what was wasted and what was missed.

Two benchmark forecasters ran alongside so the network had to earn its seat.

Total cost of being wrong per simulated store-year: wasted units at cost of goods plus missed sales at lost margin. 364 days.



PROOF OF CONCEPT

Less waste at the same availability, or much less waste at slightly lower availability

HELD-OUT YEAR, ONE STORE	WASTE (RETAIL VALUE)	WASTE, % OF PRODUCTION	AVAILABILITY (FILL RATE)	TOTAL ECONOMIC COST
Status quo (par sheet)	$138,462	18.8%	98.1%	$56,011
Forecast, availability held	$87,351 **(−37%)**	12.8%	97.6%	$41,078
Forecast, profit-optimal	$55,883 **(−60%)**	8.8%	95.5%	$39,723
Oracle (perfect knowledge)	$57,435	8.9%	96.6%	$35,788

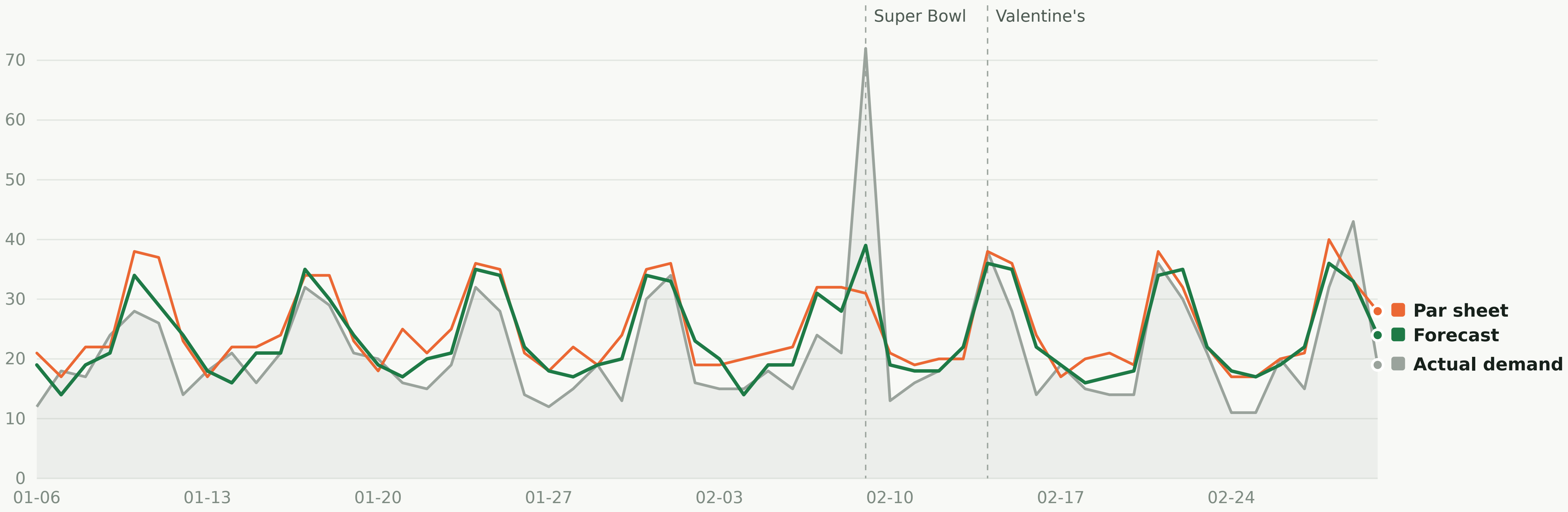
Availability held is the pilot setting: $51,111 less food thrown away per year at retail value, with customers seeing an empty case about as often as they do today.

Profit-optimal is where the decision rule would go on its own: waste falls by 60%, and availability slips from 98.1% to 95.5%. That is a business choice, item by item, not a model setting.

Forecast accuracy 13.0% weighted error, against 15.5% for a trailing average and 15.1% for a linear model.

WHAT IT LOOKS LIKE DAY TO DAY

Whole pizza, eight simulated weeks: the forecast hugs demand, the par sheet pads it



Units per day. The gray band is the demand no method ever sees; both policies only see sales. The dashed lines mark holidays the forecast knew about a day ahead.

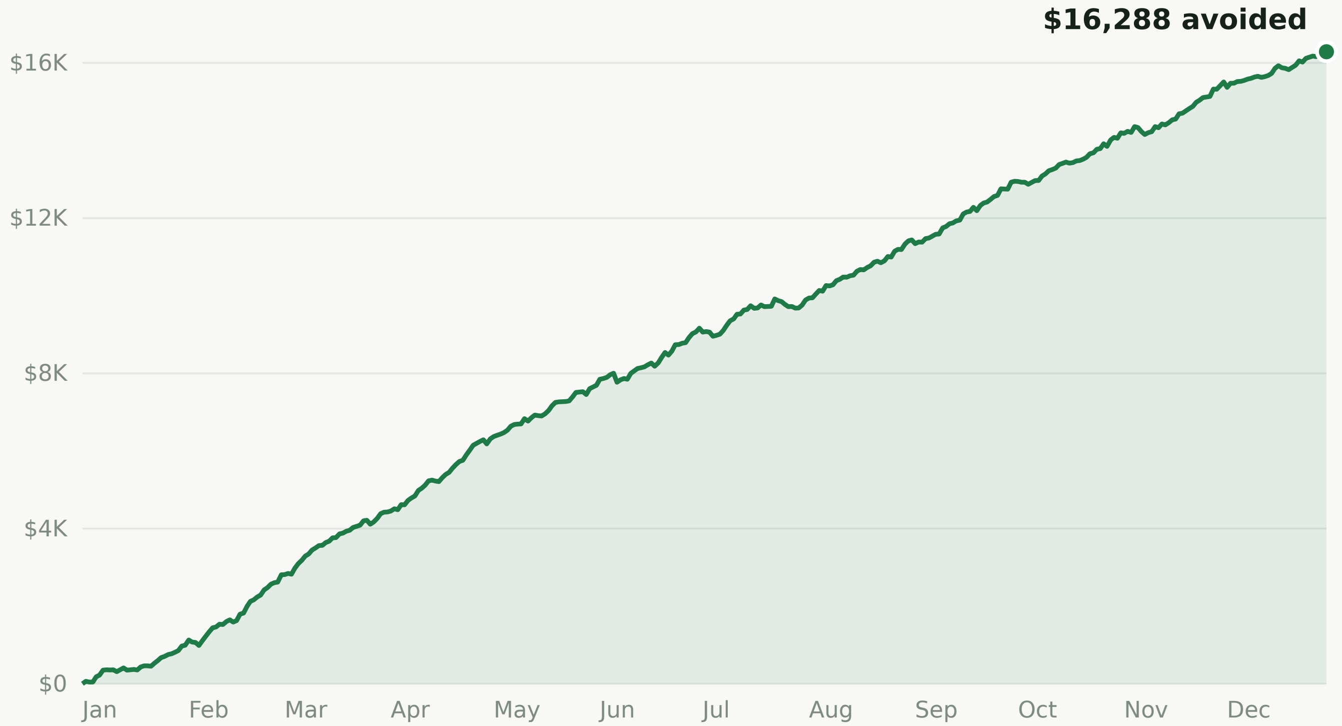
PROOF OF CONCEPT

Savings accrue every week, not at the end of a project

Cumulative economic cost avoided by the profit-optimal forecast versus the par sheet across the simulated year: $16,288 per store, or about $313 a week from the first week of live operation.

At the availability-held setting it is $14,933 per store-year in economic cost, and $51,111 of food, at retail, not thrown away.

Economic cost counts waste at cost of goods and missed sales at lost margin. Retail value of waste is the number a shrink report shows.



READ THIS SLIDE CAREFULLY

What the simulation proves, and what it does not

It proves the method

- Demand with realistic weekday, weather, holiday and seasonal structure is learnable from sales data alone, censored by sellouts as a real store's data is.
- Production driven by those forecasts beats a well-run par sheet by a wide margin, and beats two simpler forecasters too.
- Every number is reproducible from the repository with one command, and the inputs are frozen.

It does not prove our dollar figure

- The simulated store is a plausible invention. Its simulated manager, though deliberately competent, is an invention too.
- No company data of any kind was used. Nothing here has seen a real export yet.
- That is why this proposal promises 10–15% in a real pilot, not 37–60%, and why the pilot's measured baseline, not the simulation, is the number it is judged against.

THE PROPOSAL

A 90-day pilot at one store, with a stop gate before anything in the kitchen changes

WEEKS 1–2

Baseline

Log every discarded prepared-food item across 5–10 pilot SKUs with the phone-based waste logger that already exists.

Output: this store's real waste number, in dollars per year.

WEEKS 3–6

Shadow mode

With data access granted, the model retrains on this store's history and prints its morning sheet. The kitchen ignores it. We record what it would have done.

Output: forecast accuracy against reality, and five go/no-go criteria fixed in advance.

WEEKS 7–13

Live

The kitchen follows the sheet. The manager can override any line on any day, and overrides are logged, not fought.

Output: measured waste and sellout change versus the baseline.

Success criterion, agreed in advance: at least a 10% reduction in prepared-food waste dollars versus the baseline, with no material drop in on-shelf availability. If shadow mode shows the forecasts are not accurate enough, we stop at week 6 and nothing in the kitchen has changed.

THE ASKS

Three decisions, all reversible

1

Data access, in writing

Read-only access to daily sales and production counts for the pilot items at one store, for the duration of the pilot. Scoped, revocable, and requested before any real data is touched.

2

40–80 paid hours

Non-floor time to connect the store's export, run shadow mode, and operate the live phase. The importer, validator, training and scoring tools are already built and rehearsed on a mock export.

3

A sponsor and a recognition structure

A store or district leader who owns the pilot, and an agreed, written structure tied to the measured result, settled before the live phase, so that a successful pilot has somewhere to scale.

Ownership of the work built so far was deliberately kept clean: nothing was built on company data, so IP and data questions can be settled on the way in rather than argued about on the way out.

IF IT WORKS

One model serves every item, so the pilot is also the template

The same architecture scales from 9 SKUs at one store to every prepared-foods case in a district without redesign: retraining per store takes about a minute, and the morning sheet is the only interface the kitchen ever sees.

The table is arithmetic on the simulated store, not a forecast of Harris Teeter's numbers. Its job is to show which column the pilot replaces with a measured fact. The last column is what this proposal actually promises, applied to a store whose waste looks like the simulated one.

PER YEAR	FOOD NOT WASTED (RETAIL, SIMULATED)	ECONOMIC COST AVOIDED (SIMULATED)	PROMISE APPLIED (10–15% OF WASTE)
1 store	$51,111	$14,933	$13,846 – $20,769
20 stores	$1,022,216	$298,664	$276,924 – $415,386
50 stores	$2,555,539	$746,660	$692,310 – $1,038,466
100 stores	$5,111,078	$1,493,320	$1,384,621 – $2,076,931

Simulated store: $138,462 of prepared-food waste at retail per year under the par sheet; availability-held forecast setting. Replace the first row with the pilot's measured baseline and the rest follows.

RISKS, NAMED

Every risk has a gate, a fallback, or a written answer

RISK	MITIGATION
Forecasts do not transfer from simulation to this store	Shadow mode is the gate: four weeks of predictions scored against reality before production changes at all.
Availability suffers and customers notice	The pilot runs at the availability-held setting; sellouts are tracked daily and any item can revert to its par instantly.
The kitchen does not trust the sheet	Manager override is a feature, logged not fought; the pilot measures the blend, which is how it would really run.
The store's export is dirtier than expected	The importer has been rehearsed against a deliberately dirty mock export. The first real one will still surprise us, and the hours ask covers that.
Data access or IP questions	Nothing was built on company data; access is scoped, read-only, written and revocable. IP and recognition are settled in writing before the live phase.

THE DECISION

Approve one pilot store.

Week 1 begins the day the data-access letter is signed. By week 6 we know whether the forecasts are accurate here. By week 13 we have a measured number.

- Downside: a few pallets of mis-produced food, after a stop gate
- Upside: a measured template for every store

APPENDIX

Per item, simulated held-out year, profit-optimal setting

ITEM	WASTE TODAY (RETAIL)	WASTE % OF PRODUCTION	WASTE WITH FORECAST	CHANGE	FILL RATE TODAY → FORECAST	SERVICE TARGET
Hot Bar (per lb)	$36,056	18%	$11,732	−67%	98.4% → 95.5%	0.65
Rotisserie Chicken	$24,002	19%	$8,094	−66%	98.8% → 95.2%	0.60
Sushi Roll	$21,028	17%	$5,448	−74%	97.1% → 89.2%	0.58
Whole Pizza	$12,544	18%	$7,023	−44%	97.3% → 95.7%	0.68
Pizza Slice	$10,976	22%	$4,837	−56%	98.6% → 96.1%	0.74
Sub / Sandwich	$9,776	19%	$5,607	−43%	97.4% → 96.4%	0.70
Doughnut	$8,727	19%	$4,612	−47%	98.7% → 97.3%	0.79
Bread Loaf	$8,092	18%	$4,229	−48%	98.1% → 96.7%	0.72
Cake	$7,261	25%	$4,300	−41%	93.1% → 91.8%	0.65

Service target is the newsvendor critical fractile each item's price, cost and salvage imply: the fraction of days the item should not sell out. Expensive, low-margin items get a lower target and run lean; cheap, high-margin items pad.

APPENDIX

Method notes and provenance

Demand model

A global quantile-regression neural network: a GRU encoder over each item's trailing 28 days, learned item embeddings, and calendar, weather and holiday covariates, trained with pinball loss across 11 quantiles. Sellout days are treated as censored, since sales only bound demand from below when the case sold out.

Decision layer

Per-item newsvendor critical fractile, rounded to batch sizes. The availability-held variant raises each item's target until the year's fill rate matches the par sheet's.

Validation

Trained on simulated 2023–24, tested untouched on 2025 against naive, linear and oracle policies. Weighted forecast error: network 13.0%, trailing average 15.5%, linear 15.1%.

Provenance

The synthetic dataset, the trained checkpoint and the results file are frozen and checked in. `python -m model.backtest` reproduces the results byte for byte, and the tools refuse to overwrite the frozen files by accident. This deck is generated from that results file by `tools/build_deck.py`.

Real-data path

Importer, validation gate, training, evaluation and the morning-sheet loop have been run end to end on a mock export built from the synthetic store with realistic dirt added. None of it has seen real data.